# PROJECT PJ: A COGNITIVE AMPLIFIER FOR PROFESSIONAL SWARMS

Integrating Multi-Agent Cascades, Model Context Protocol, and Cryptographic Consensus

**Kaggle 5-Day Intensive: AI Agents Capstone Project (June 15 - 19, 2026)**

**Architect & Author: Devs One — Danny Bouldiez**

**Platform: Google AI Studio, Google Antigravity & Nous Research Hermes Agent**

## ■ Executive Summary

Conventional enterprise workflows scale horizontally by adding headcount—introducing communication overhead, scheduling delays, and structural coordination errors. **Project PJ (PodJobs.ai)** proves a new paradigm of **Cognitive Amplification**: *one human specialist directing a synchronized, parallel consensus department of 12 intelligent AI agents.*

By combining the **Google GenAI SDK**, **Model Context Protocol (MCP)**, **NVIDIA NeMo Guardrails emulation**, and **cryptographic Merkle Root consensus signatures**, Project PJ provides a secure, zero-overhead workflow sandbox that scales capabilities instead of headcount.

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## ■■ The Challenge & The Solution

| The Corporate Problem | Project PJ Solution |
|---|---|
| **Execution Latency**: 3-10 days for complex documents. | **Asynchronous Swarms**: Multi-agent tasks finished in under 15 seconds. |
| **Verification Overhead**: Peer-reviews introduce delays. | **Bayesian Consensus**: Unanimous 12-node voting verification. |
| **Security Risks**: Confidentiality leaks and input injection. | **Isolation Shield**: Prompt sanitization and NeMo Guardrail thresholds. |
| **Audit Trails**: Hard to track steps in multi-agent reasoning. | **Merkle-Attestation Seals**: Immutable cryptographic signatures. |

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## ■■ Architecture & Sequential Cascade (ADK)

Project PJ implements a 5-node sequential execution cascade utilizing the **Google GenAI SDK** (**`gemini-3.5-flash`**):

```
graph TD
A[Human Director Prompt] --> B[Planner Node]
B -->|Deconstructs goals| C[Context Miner RAG Node]
C -->|Mines case files / documents| D[Suite Draftsman Node]
D -->|Synthesizes professional briefs| E[System Integrity Auditor Node]
E -->|Scans for safety & biases| F[Swarms Consensus Arbiter Node]
F -->|Bayesian consensus seal| G[Merkle Root Cryptographic Seal]
```

1. **Planner Node**: Maps out execution branches and allocates agent responsibilities.

2. **Context Miner Node**: Performs vector grounding (RAG) based on local knowledge databases.

3. **Suite Draftsman Node**: Compiles professional documents and templates.

4. **System Integrity Auditor Node**: Emulates **NVIDIA NeMo Guardrails**, rejecting any payload with a bias/safety score exceeding `0.05`.

5. **Swarms Consensus Arbiter Node**: Performs final validation and returns the output along with a mathematical seal.

### Cryptographic Consensus & Merkle Attestation

To guarantee the logical integrity of the reasoning path, the input, actions, and outputs of all agent nodes are sequentially hashed and sealed into a **Merkle Tree**. The resulting **Merkle Root Hash** acts as an immutable tamper-proof signature, proving that no node's intermediate outputs were modified during the cascade.

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## ■ Local CLI & MCP Extensibility

To enable external developer tools and shell automation, Project PJ exposes:

- **JSON-RPC MCP Server (`mcp-server/index.js`)**: A stdio-based server compliant with the Model Context Protocol, exposing swarm simulation and inspect tools to IDEs like Claude Desktop.
- **Command-Line Agent Skill (`bin/podjobs-cli.js`)**: A CLI enabling offline simulations and live sequential runs directly from the developer shell.
- **Deployed API Validator (`bin/validate-live-api.js`)**: A testing harness verifying that Vercel serverless routes correctly connect with live Google APIs.

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## ■ The Nous Research Hermes Connection

Project PJ integrates the state-of-the-art **Nous Research Hermes Agent** configuration framework. By utilizing Nous Research blueprints, each node loaded in the dynamic workstation utilizes customized `hermes.json` onboarding manifests, configuring:

- Strategic organizational directives.

- Persona alignment parameters.
- Isolated model instruction sets.

### License Credit & Attestation

We extend our deepest gratitude to the **Nous Research** team for their contributions to open-source agent architectures. Nous Research Hermes and its derivative configurations are utilized in this project under the terms of the Apache License 2.0.



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## ■ Attestation & Sign-off

*"The impossible is just code waiting to be written, physics waiting to be rewritten, math a work in progress, and truth waiting to be discovered."*

>

— ***Devs One*** *(Danny Bouldiez)*

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